

AI-Powered Credit Risk Intelligence Platform

Explainable risk scoring and governed natural-language analytics

Yash Barjatya

Home Credit application data

One workspace connects evidence, prediction and governed analytics



Talk-to-data path

Question → bounded context → one LLM SQL proposal → parser and allowlist → read-only SQLite → deterministic answer

The LLM never calculates applicant risk. Prediction and analytics remain separate, inspectable paths.

The outcome is rare, so average precision matters

307,511

applications

8.07%

payment-difficulty outcomes

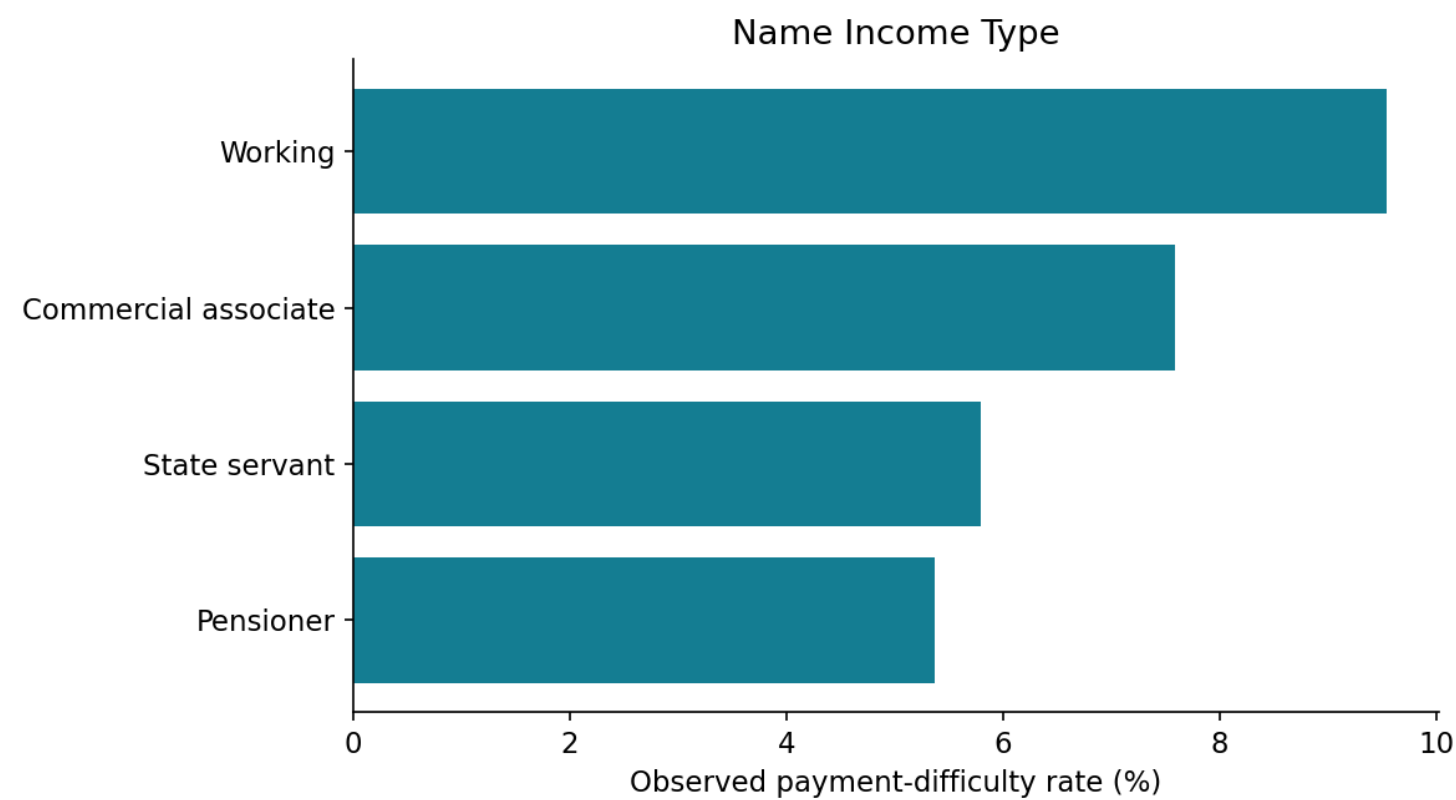
TARGET means late payment beyond an unspecified delay in early installments.

It is not a universal legal default definition. Amount currency is not asserted because the supplied schema does not establish it.

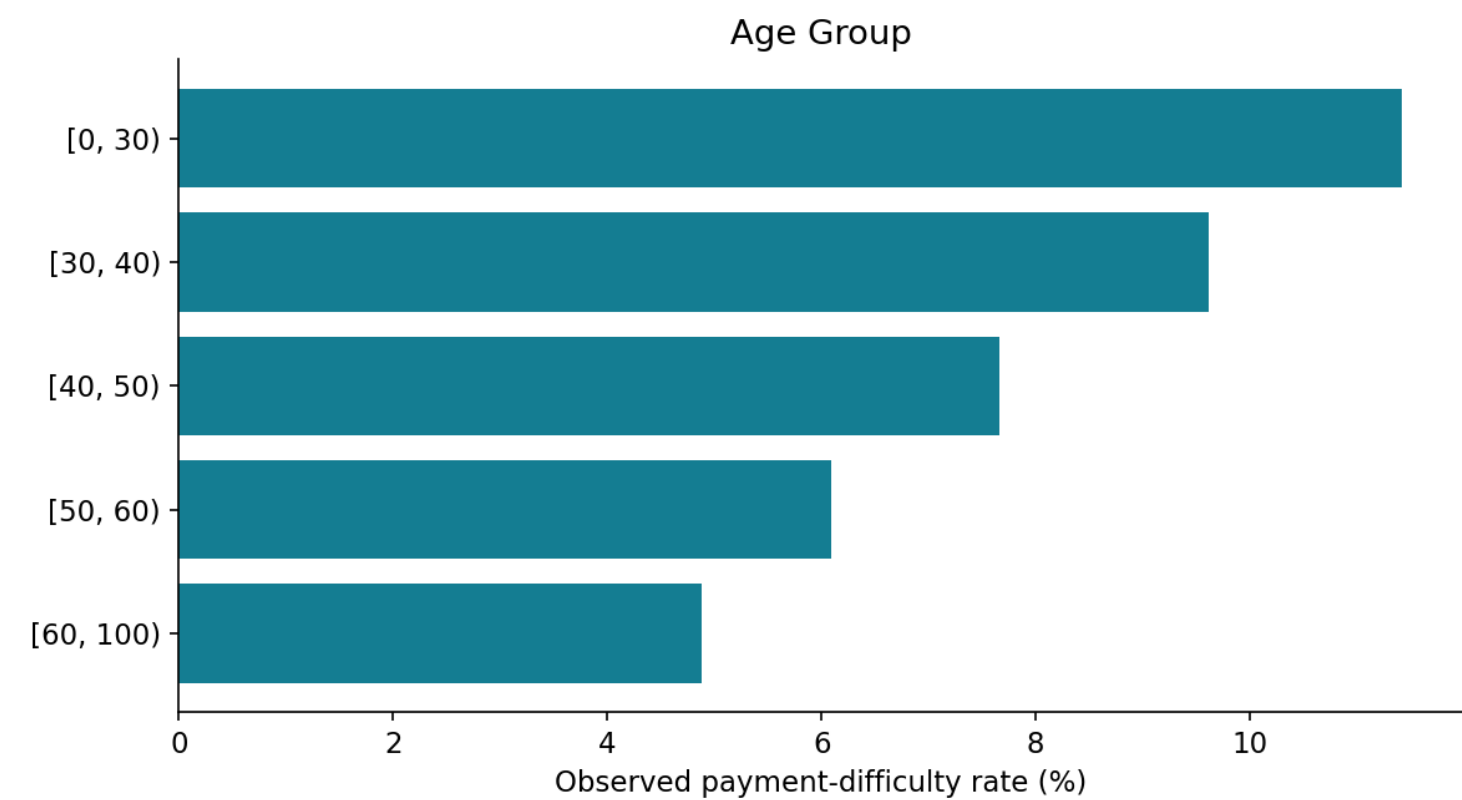
Data-quality decisions

- Final test rows excluded from EDA and selection
- DAYS_EMPLOYED=365243 treated as a sentinel
- Unknown categories accepted; medians learned on training rows
- Target and applicant ID excluded from features
- Unavailable detailed bureau/repayment tables disclosed

Six observed group differences guide investigation - not decisions

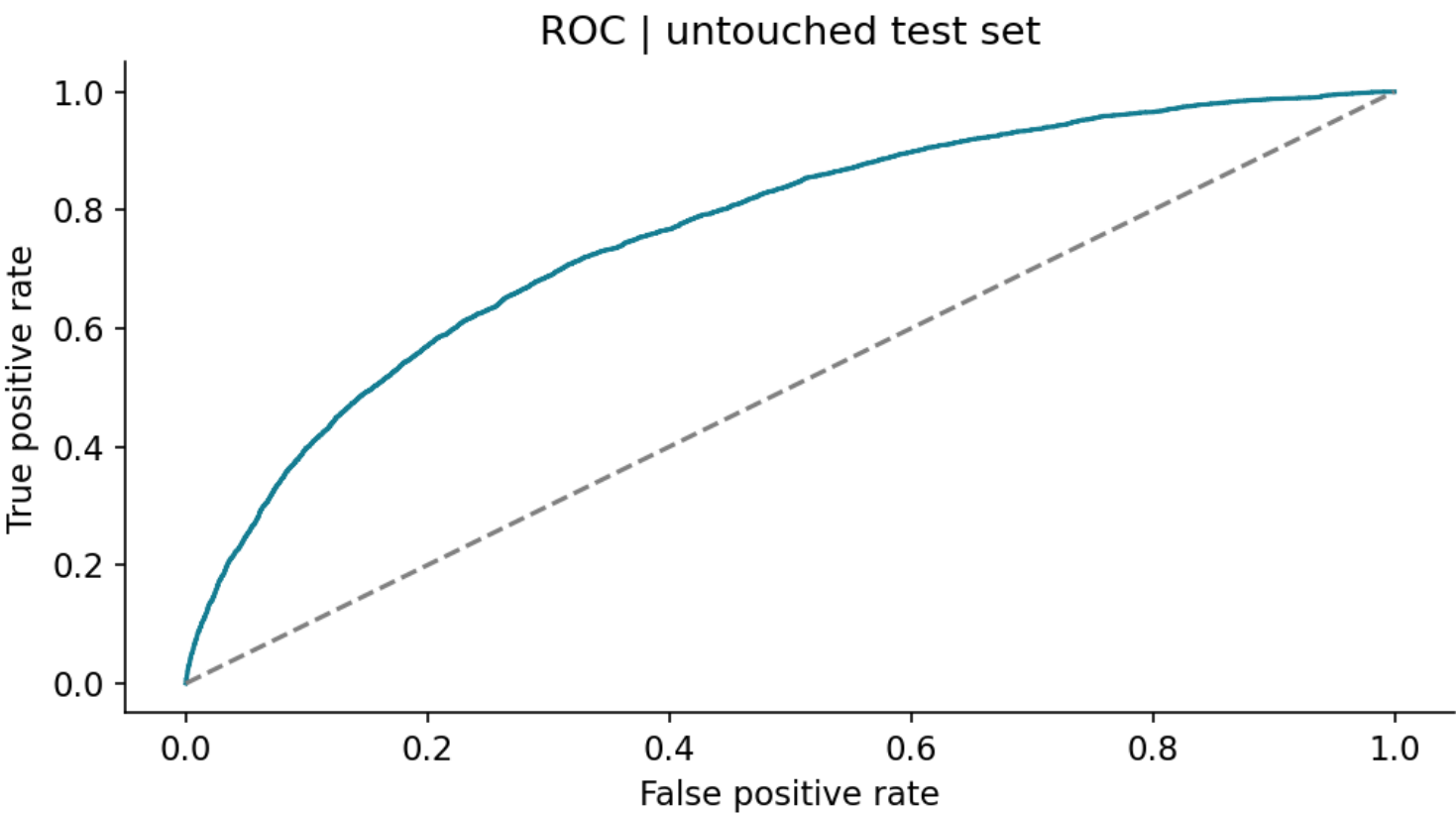


Working: 9.5% vs Pensioner: 5.4% (development data)



Under 30: 11.4% vs age 60-99: 4.9% (development data)

Calibrated boosting improves ranking over the baseline

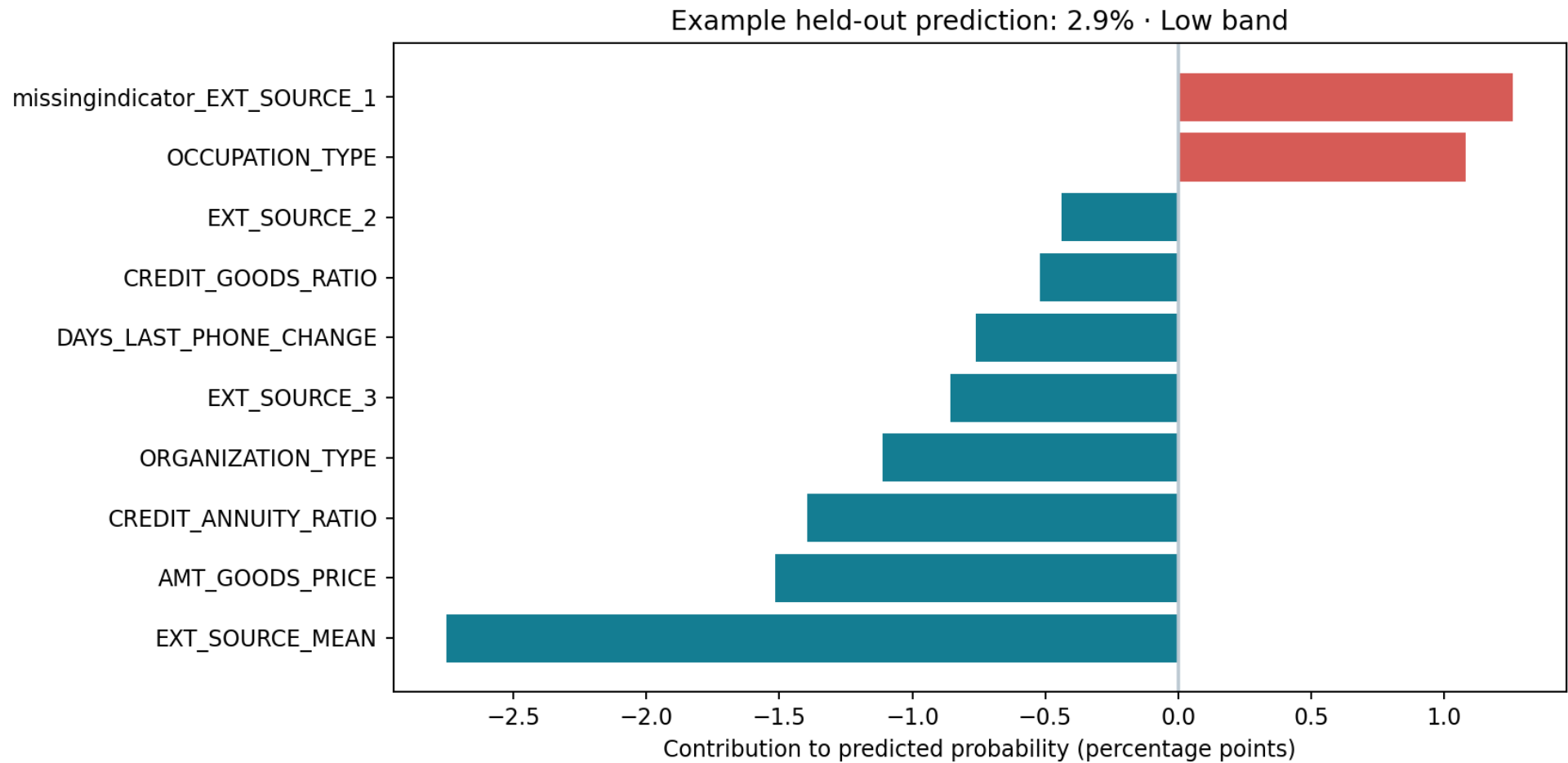


FINAL TEST

ROC-AUC	0.7616
Average precision	0.2438
Recall at 0.08	70.19%
Precision at 0.08	16.50%
Brier score	0.06777

The F2-selected threshold prioritizes recall and creates many false positives. It is an experimental screening threshold, not lending policy.

Risk bands and SHAP make one estimate readable



HELD-OUT EXAMPLE

2.88 / 100

estimated risk score

LOW

relative validation band

SHAP contributions reconstruct the calibrated probability. Correlated inputs may share attribution; values are explanatory, not causal.

A compact rule surrogate agrees with model bands 74.73% of the time

High

Mean external score ≤ 0.34 and age ≤ 56.35

Medium

Mean external score > 0.34 and ≤ 0.53

Low

Mean external score > 0.53

Depth-three tree fitted on training model predictions. Exact saved tree and imputation control outputs. These rules are an approximation, not policy.

The chatbot exposes evidence and refuses unsafe query shapes

Five reference patterns

1. Overall development difficulty rate
2. Rates by income type
3. Positive-case average credit
4. Age-decade comparison
5. Occupation ranking with minimum n=500

Hallucination controls

- One SELECT; one allowed table
- Known columns and functions only
- No joins, subqueries, CTEs or mutations
- 100-row cap + 2-second SQLite budget
- Local numeric summary; SQL and rows visible

Live Gemini-to-SQL validation succeeded after Docker startup

A natural-language question produced validated SQL, returned calculated rows and reported token usage. The five reference queries also run directly against SQLite without a provider call.

Docker provides the same reproducible path locally and in deployment

1

Mount CSV

private, read-only

2

Build image

Python 3.12

3

Bootstrap

SQLite + profiles

4

Serve

Streamlit :8501

Included

Dockerfile · docker-compose.yml · .env.example · non-root user · health check · setup instructions

Public deployment

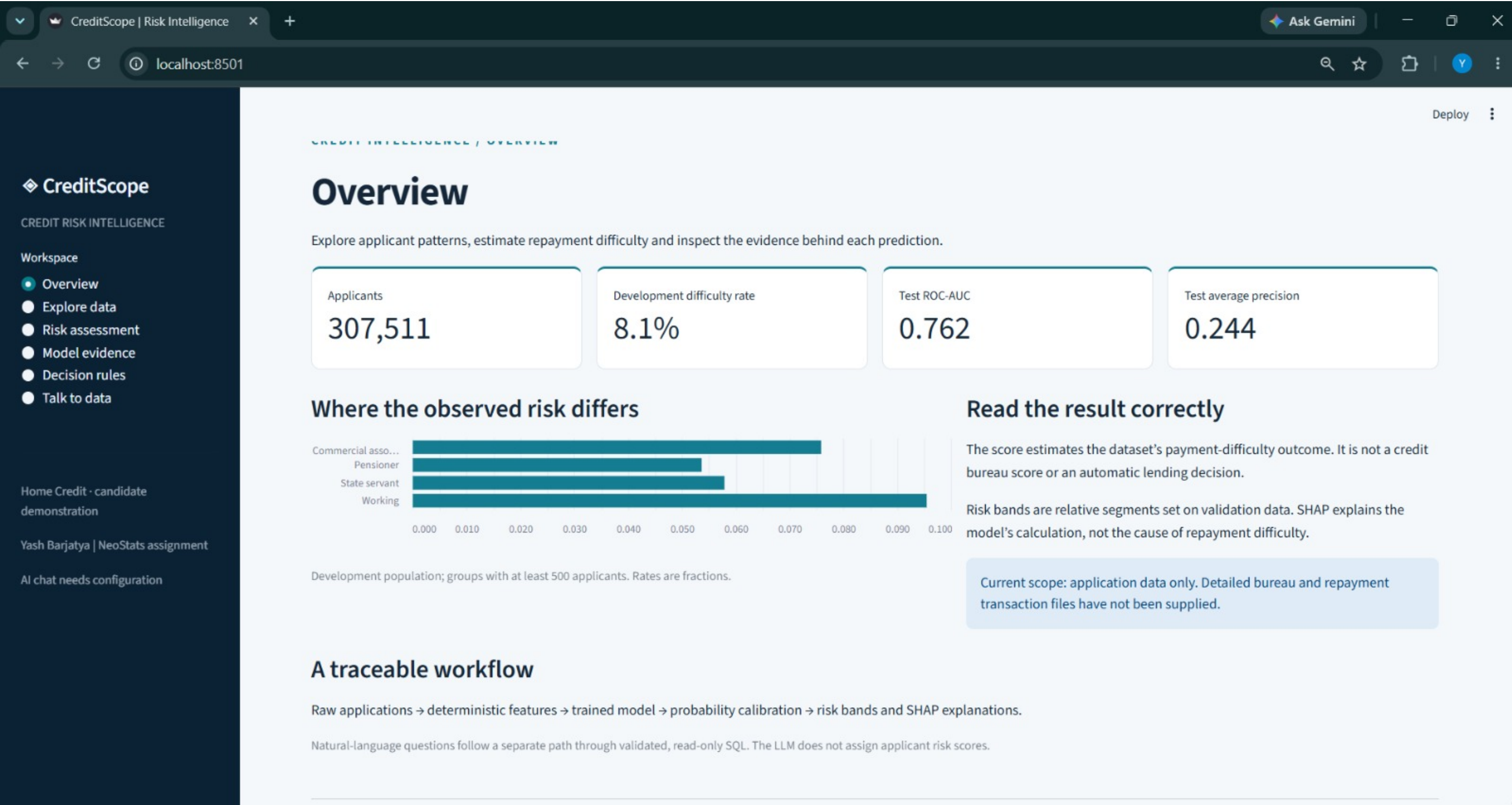
The Docker web service is available at credit-risk-intelligence-platform-1.onrender.com. Raw applicant data and API secrets remain outside the public repository.

Ready for technical review

01	Docker + tests	Healthy container; 13 tests passed
02	Live LLM	Gemini proposed validated SQL and returned results
03	Evidence	Application outputs included in this PDF
04	Public URL	credit-risk-intelligence-platform-1.onrender.com

Next improvements: historical tables, time-based/external validation, fairness analysis, cost-based threshold and production authentication.

Live workspace overview



EDA: observed rates and business findings

CreditScope

CREDIT RISK INTELLIGENCE

Workspace

Overview

Explore data

Risk assessment

Model evidence

Decision rules

Talk to data

Home Credit · candidate demonstration

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AI chat needs configuration

Deploy

Explore data

Insights use development rows only, keeping the final holdout separate during model development.

Compare applicant groups

NameIncomeType

Commercial asso...

Pensioner

State servant

Working

0.00

0.01

0.02

0.03

0.04

0.05

0.06

0.07

0.08

0.09

0.10

Group	Applicants	Difficulty cases	Difficulty rate
Working	134939	12881	0.0955
Commercial associate	60993	4626	0.0758
State servant	18535	1073	0.0579
Pensioner	46872	2515	0.0537

Six findings from the data

1. NAME_INCOME_TYPE: Working has a 9.5% payment-difficulty rate (n=134,939), compared with 5.4% for Pensioner (n=46,872). Association, not causation.

2. NAME_EDUCATION_TYPE: Lower secondary has a 10.9% payment-difficulty rate (n=3,225), compared with 5.4% for Higher education (n=63,584). Association, not causation.

3. NAME_HOUSING_TYPE: Rented apartment has a 12.2% payment-difficulty rate (n=4,136), compared with 6.6% for Office apartment (n=2,227). Association, not causation.

Risk assessment: probability, score and band

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Risk assessment

Input source

Manual scenario

Held-out applicant

An editable hypothetical scenario, not a real applicant. Unprovided fields use training preprocessing defaults.

Annual income (dataset units)

180000.00

-

+

Credit amount (dataset units)

600000.00

-

+

Annuity (dataset units)

30000.00

-

+

Age in years

35

-

+

Income type

Working

Education

Secondary / secondary special

Ext Source 1 (optional)

-

+

Ext Source 2 (optional)

-

+

Ext Source 3 (optional)

-

+

> Remaining input fields

Assess repayment risk

Results from the last submitted assessment; submit again after editing.

Estimated difficulty probability

12.4%

Risk score / 100

12.4

Relative risk band

Medium

Missing numeric features

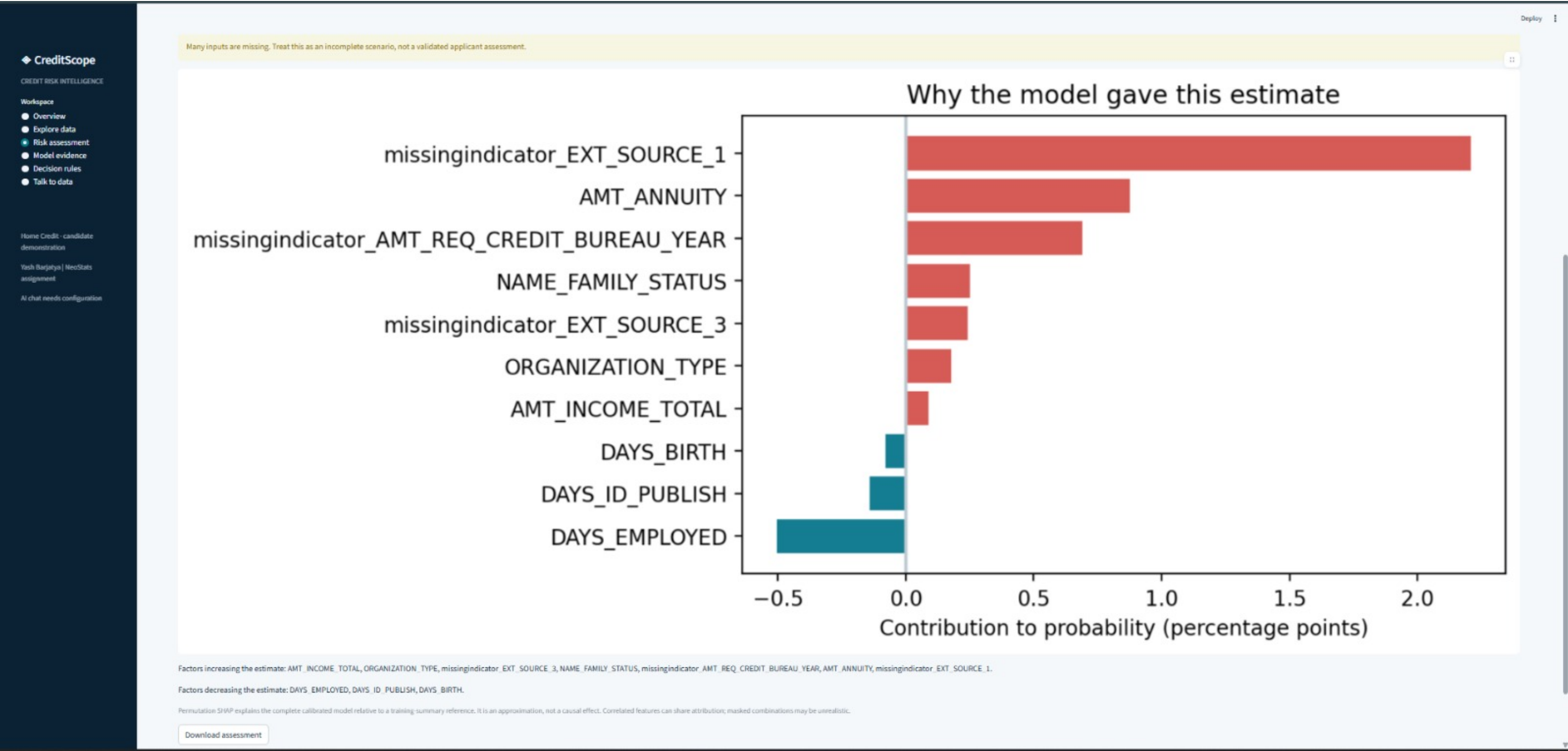
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Many inputs are missing. Treat this as an incomplete scenario, not a validated applicant assessment.

Captured from the local Docker run; pages 11-17 were captured before provider configuration. Page 18 verifies the configured live Gemini flow.

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Prediction explanation: SHAP contribution view



Model evidence: split discipline and test curves

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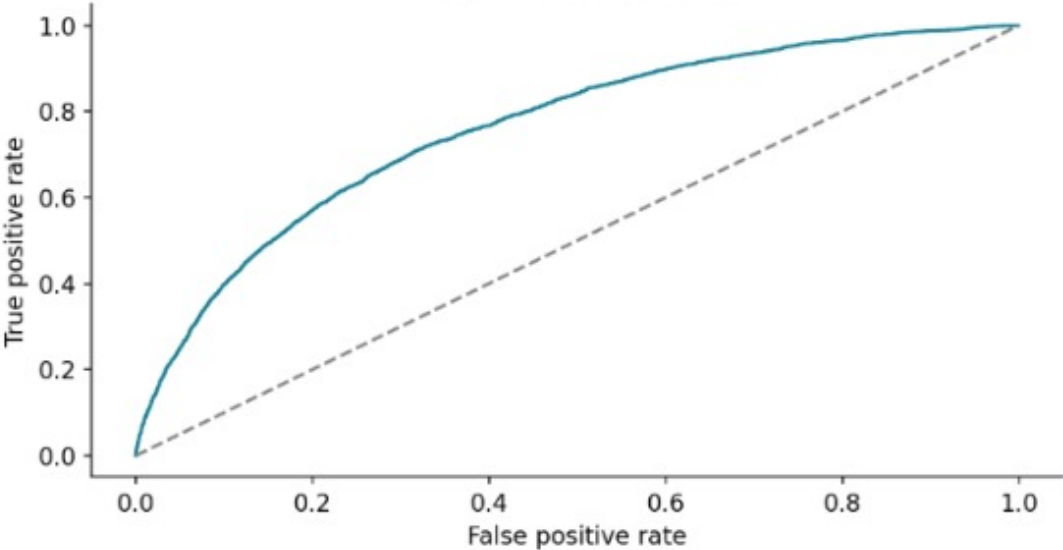
Model evidence

Selection uses validation average precision. Calibration uses a separate split; final test metrics follow these choices.

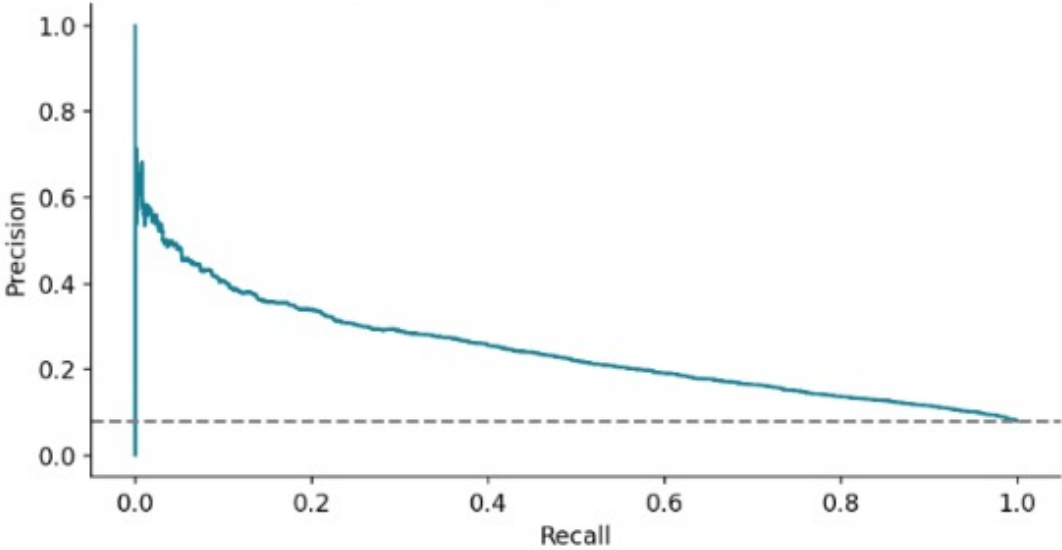
	train	calibration	validation	test
Applicants	169130	46127	46127	46127

Evaluation	roc_auc	pr_auc_average_precision	precision	recall	f1	brier	threshold
Logistic baseline · validation	0.7475	0.2274	0.1604	0.675	0.2592	0.2024	0.5
Selected model · validation	0.7583	0.242	0.1633	0.69	0.2642	0.0679	0.08
Selected model · final test	0.7616	0.2438	0.165	0.7019	0.2672	0.0678	0.08

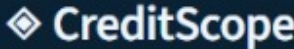
ROC | untouched test set



Precision Recall | untouched test set



Model evidence: confusion matrix and calibration



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AI chat needs configuration

0.2

0.0

0.0

0.2

0.4

0.6

0.8

1.0

False positive rate

	Screen negative	Screen positive
Actual no difficulty	29178	13225
Actual difficulty	1110	2614

0.2

0.0

0.0

0.2

0.4

0.6

0.8

1.0

Recall

Calibration | test set

0.5

0.4

0.3

0.2

0.1

0.0

0.0

0.1

0.2

0.3

0.4

0.5

Mean predicted risk

Deploy

Threshold choice: Maximize F2 on validation; recall weighted more than precision. No business cost model supplied.

Band choice: Validation probability quantiles: bottom 50% Low, next 35% Medium, top 15% High. Relative demo segments, not policy.

The threshold prioritizes recall and produces many false positives. A real credit policy needs explicit costs, external/time-based validation and fairness assessment.

This application-only model does not use detailed bureau or repayment histories.

Business-readable decision-rule surrogate

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Deploy

Validation agreement with model bands

74.7%

A depth-three tree approximates the main model's bands using five interpretable features. It is fitted on training predictions.

Examples in plain language (rounded thresholds):

• Mean external score at most 0.34 and age at most 56.35 years: the surrogate assigns High risk.

• Mean external score above 0.34 and at most 0.53: the surrogate assigns Medium risk.

• Mean external score above 0.53: the surrogate assigns Low risk.

These examples summarize fitted branches. The exact saved tree and its imputation define the rule predictions.

```
|--- EXT_SOURCE_MEAN <= 0.53
| |--- EXT_SOURCE_MEAN <= 0.34
| | |--- AGE_YEARS <= 56.35
| | | |--- class: High
| | |--- AGE_YEARS > 56.35
| | | |--- class: Medium
| |--- EXT_SOURCE_MEAN > 0.34
| | |--- EXT_SOURCE_MEAN <= 0.46
| | | |--- class: Medium
| | |--- EXT_SOURCE_MEAN > 0.46
| | | |--- class: Medium
|--- EXT_SOURCE_MEAN > 0.53
| |--- EXT_SOURCE_MEAN <= 0.58
| | |--- EMPLOYED_YEARS <= 4.50
| | | |--- class: Low
| | |--- EMPLOYED_YEARS > 4.50
| | | |--- class: Low
| |--- EXT_SOURCE_MEAN > 0.58
| | |--- EMPLOYED_YEARS <= 4.45
| | | |--- class: Low
| | |--- EMPLOYED_YEARS > 4.45
| | | |--- class: Low
```

Approximation of model risk bands; not lending policy or causal rules.

Live Gemini-to-SQL answer with evidence



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● **Talk to data**

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AI chat configured

Talk to data

Ask about observed patterns. Inspect SQL and evidence with every answer.

Clear conversation

> Five verified SQL examples (no AI call)

 Which income types have the highest payment difficulty rates?

 The query returned 4 groups or records. The table shows the calculated results in SQL order. Rate columns are fractions (0.08 means 8%). Results are capped at 100 rows.

```
SELECT income_type, COUNT(*) AS applicants, AVG(payment_difficulty) AS difficulty_rate FROM applicants WHERE split <> 'test' GROUP BY income_type HAVING COUNT(*) >= 500 ORDER BY difficulty_rate DESC LIMIT 100
```

income_type	applicants	difficulty_rate
Working	134939	0.0955
Commercial associate	60993	0.0758
State servant	18535	0.0579
Pensioner	46872	0.0537

Provider-reported token usage: {'completion_tokens': 64, 'prompt_tokens': 465, 'total_tokens': 529}

Only schema, question and three prior question/SQL pairs go to the provider. Results are summarized locally, without sending applicant rows or inventing numerical conclusions.

Candidate demonstration · application data only · decision support, not loan approval.

Which income types have higher difficulty rates?

Captured after provider configuration: live Gemini → validated SQL → calculated rows.

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